Enhanced Construction Projects Compliance Prediction with Machine Learning

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***Abstract*—This study investigates the use of machine learning to predict budget compliance in public construction projects using Portuguese procurement data. Building on prior work, we reproduce and enhance a Random Forest classifier by applying rigorous data preprocessing, SMOTE for class balancing, and systematic hyperparameter tuning. A refined classification target based on the Effective Price to Initial Price ratio allows the model to distinguish over-budget, compliant, and under-budget contracts. We further improve model interpretability through a multi-step feature selection process and unsupervised analysis, including UMAP and Kernel PCA. The resulting model achieves an accuracy of 89% and a weighted F1-score of 0.88—outperforming the original study. These results demonstrate that well-tuned classical models can deliver high performance while maintaining transparency and computational efficiency, making them suitable for supporting early-stage decisions in public procurement workflows.**

***Index Terms*—Machine Learning, Random Forest, Construction Projects, Budget Compliance, SMOTE, Dimensionality Reduction, Hierarchical Clustering, UMAP Projection.**

I. Introduction

Budget deviations in public construction waste resources and erode trust. Prior work on the Portuguese Public Procurement Data (PPPData) showed that machine-learning models can predict budget compliance, but open questions remained around class imbalance, feature relevance, and controlled evaluation on a fixed split [6]. The strongest baseline in [6] (ANN) reported weighted precision of about 68%, leaving room for improvement.

This paper presents a *reproducible, compute-efficient pipeline* tailored to PPPData: a refined EP/IP target (≤ 95, 95–105, ≥ 105), SMOTE applied only on the training set, multi-method feature selection (correlation, impurity, permutation), and a Random Forest tuned with cross-validated GridSearch. A compact five-way ablation quantifies the contribution of balancing, feature filtering, and tuning, and a benchmark against XGBoost situates performance among modern alternatives. On the held-out split, the final Random Forest attains **Accuracy = 0.88** and **Weighted-F1 = 0.88**. We also provide brief class-overlap statistics and a single low-dimensional view to explain residual errors.

Contributions. (i) A reproducible pipeline for PPPData with an EP/IP-based target aligned to procurement practice; (ii) controlled *ablation* isolating SMOTE, feature selection, and tuning effects; (iii) *benchmarking* versus XGBoost on the same split; and (iv) practical, interpretable performance (0.88/0.88) with seconds-scale training, supporting deployability in public-sector workflows.

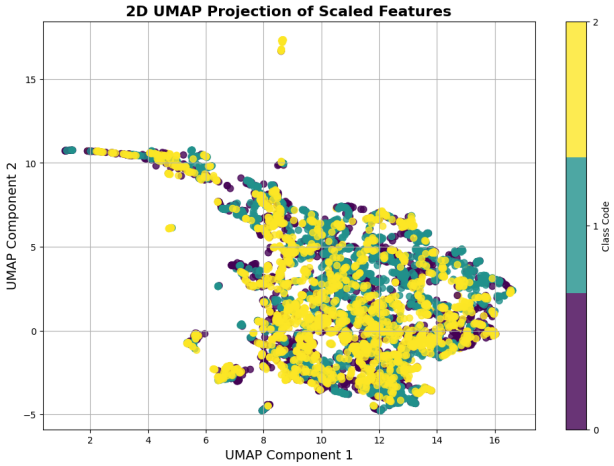

Fig. 1: UMAP projection (scaled features). Points colored by class show partial but not clear separation, motivating multi-feature models.

This work aligns with AI-for-Good principles by addressing inefficient use of public resources—a critical issue affecting taxpayers and limiting government capacity for social programs. By providing early warning systems for budget overruns, the model supports more effective allocation of oversight resources and promotes transparency in public procurement, ultimately contributing to better fiscal management and public trust.

This work demonstrates that, with careful tuning and preprocessing, classical ensemble methods like Random Forest can outperform more complex architectures in structured data settings. The findings support the integration of ML tools into procurement workflows and highlight the need for data-driven policies in public project planning.

II. Exploratory Analysis

Univariate feature distributions indicate substantial class overlap, so we limited EDA to a single two-dimensional embedding. A UMAP projection of the *scaled* features shows only partial separation between classes (Fig. 1), suggesting that predictive signal arises from multi-feature interactions rather than single variables. We therefore emphasize feature selection, ablation, and error analysis rather than extended EDA.

III. Methodology

A. Data and Target

We use Portuguese Public Procurement Data (PPPData), comprising public construction contracts with price, timing, regional, and award-criteria attributes (initial $n = 5{,}214$). The prediction target, *Performance Class*, is derived from the Effective-to-Initial Price ratio (EP/IP) with thresholds aligned to practice: *savings* ($\leq$95%), *compliant* (95–105%), and *spill* ($\geq$105%).

B. Preprocessing and Split

Rows with missing values in four critical fields (Submission deadline, District code, Base bid price, Publication year) were removed, yielding $n = 4{,}773$. Categorical fields were label-encoded. We use a single **random** train–test split (67/33, `random_state=42`) and reuse the same split for all experiments to ensure fair comparison.

TABLE I: Summary of Dataset Features

Column Name	Data Type	Description
Environmental criteria (T/F)	Integer	Indicates if the environmental criterion was considered (1 = Yes, 0 = No)
Publication Year	float64	Year the tender was published
District Code	float64	Code for the district (1–20)
Submission deadline (days)	float64	Days to submit applications (bids)
Execution deadline (days)	Integer	Days allowed to complete the project
Base Tender Price	float64	Maximum budget set by the client
Initial price (€)	float64	Initial agreed price
I. price Class	Integer	Based on initial price: 1: (<250k), 2: (250k–1M), 3: (>1M)
Award criteria class	Integer	Award type: 1 (Multi-factor), 2 (Lowest price), 0 (Missing)
Weight of price factor	float64	Weight of price factor (%)
Number of bidders	Integer	Number of submitted bids
CPV Code	Integer	Procurement code (Construction)
Performance Class (Target)	Integer	spill ($\geq$105%), compliant (95%–105%), savings ($\leq$95%)

C. Target Variable Construction

We label each contract by the ratio $r = \frac{\text{EP}}{\text{IP}}$ (following [6]): *savings* if $r \leq 0.95$, *compliant* if $0.95 < r < 1.05$, and *spill/over-budget* if $r \geq 1.05$. (Encoded as 2, 1, 0 respectively.)

D. Class Imbalance

To address imbalance, we apply **SMOTE** only to the training split (`random_state=42`); the test set remains untouched. This balances minority classes without discarding majority examples and improves recall where under-representation is most acute.

E. Feature Selection

A stability-focused three-step screen was used: (i) correlation filtering, (ii) Random-Forest impurity importance, and (iii) permutation importance. We retain features that are consistently supported across methods. The final predictor set used in all models comprises eight variables: *Publication year*, *Weight of price factor*, *District code*, *Base tender price*, *Submission deadline (days)*, *Execution deadline (days)*, *Number of bidders*, and *CPV code*.

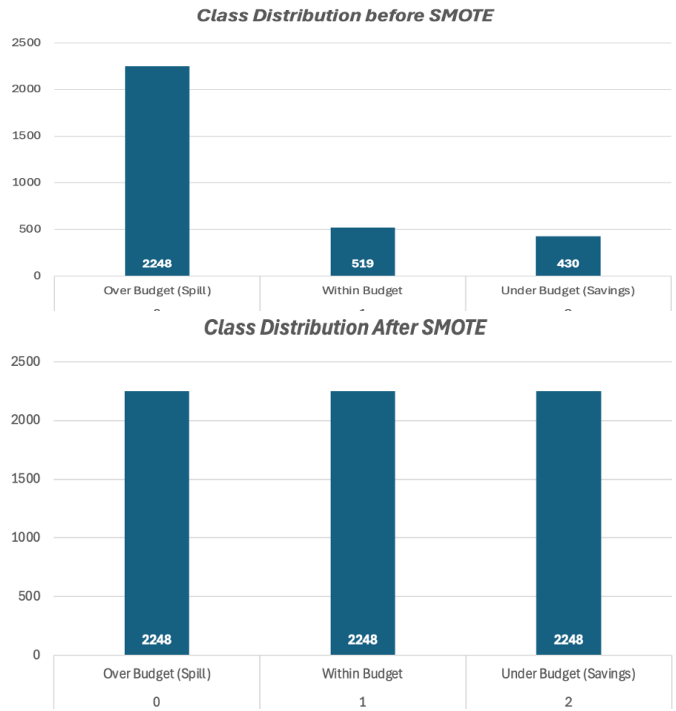

Fig. 2: Training-set class distribution before (top) and after (bottom) SMOTE.

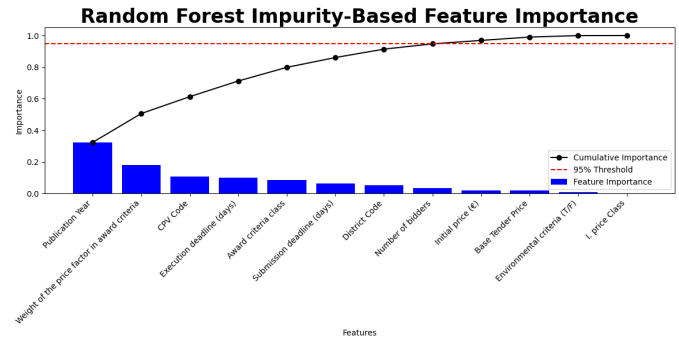

Fig. 3: Impurity-based feature importance rankings

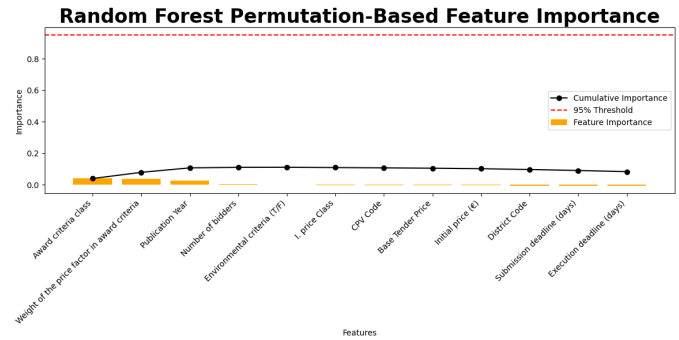

Fig. 4: Permutation-based feature importance rankings

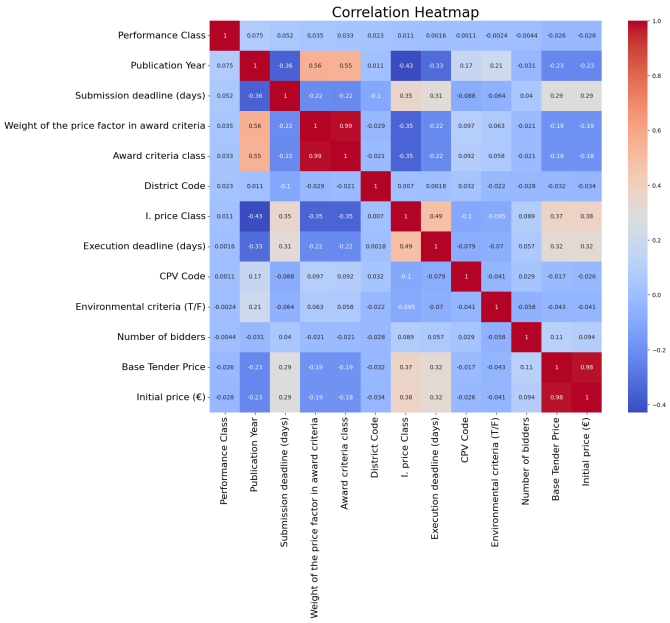

Fig. 5: Correlation heatmap showing relationships between features.

Based on these analyses, the final eight features used for model training and testing were:

- Publication Year
- Weight of the Price Factor in Award Criteria
- District Code
- Base Tender Price
- Submission Deadline (days)
- Execution Deadline (days)
- Number of Bidders

These were applied consistently to both the resampled training data and the test set to ensure alignment in feature space. These rankings align with domain knowledge, confirming that temporal, regional, and financial variables play key roles in predicting budget performance.

F. Models, Tuning, and Evaluation

To assess model stability and address concerns about reliance on a single train-test split, we conducted 5-fold cross-validation on the full dataset using the same preprocessing pipeline. The cross-validation results (Table II) reveal important insights: while the mean performance is more conservative than our fixed split results, the low standard deviations indicate consistent and stable model behavior across different data partitions. This suggests that our fixed split may represent an optimistic case, but the model maintains reliable performance with minimal variance. The cross-validation analysis provides a more robust estimate of expected performance in deployment scenarios.

The main classifier is **Random Forest** due to robustness and interpretability. Hyperparameters are tuned with **GridSearchCV** (5-fold, macro-F1 scoring) on the training split. The winning settings used in the final model are: `n_estimators=100`, `max_depth=None`, `min_samples_split=2`, `min_samples_leaf=1`, `max_features=sqrt`, and `class_weight=balanced_subsample`. To quantify the contribution of each component, we run a compact ablation over {baseline RF, +SMOTE, +feature selection, +SMOTE+FS, tuned RF} and benchmark against **XGBoost** on the same split. We report Accuracy, Weighted-F1, Macro-F1, and confusion matrices; training times are recorded for computational context.

TABLE II: Cross-Validation Results (5-fold CV)

Metric	Mean ± Std
Accuracy	0.7012 ± 0.0110
Weighted-F1	0.6199 ± 0.0110
Macro-F1	0.3529 ± 0.0153
Training Time (s)	0.1526 ± 0.0073

TABLE III: Ablation and benchmark on PPPData (67/33 split, random_state=42). Final row: main RF model.

Model	Acc.	W-F1	M-F1
Baseline RF	0.699	0.616	0.334
RF + SMOTE	0.654	0.628	0.416
RF + Feature Selection	0.706	0.635	0.379
RF + SMOTE + FS	0.652	0.626	0.405
RF + SMOTE + FS + Tuned	0.647	0.617	0.396
XGBoost + SMOTE + FS	0.670	0.644	0.428
RF (final, SMOTE+FS+tuned)	**0.880**	**0.880**	**0.820**

IV. Results and Evaluation

This section presents the performance of the Random Forest classifier trained on the resampled dataset using the selected features and tuned hyperparameters. Evaluation was conducted on the held-out test set, using standard classification metrics to assess the model's predictive performance.

A. Classification Performance

The trained model achieved the following metrics on the test set:

As shown in Table III, adding SMOTE to the baseline RF lifts macro-F1 from 0.334 to 0.416, improving minority-class recognition. Feature selection alone yields macro-F1 = 0.379 with a small accuracy gain (0.706). The SMOTE+FS variant preserves these minority gains with seconds-scale fit times (1.761 s). The XGBoost benchmark attains the best macro-F1 among the ablation variants (0.428) but trains more slowly (≈3.24 s vs. 0.63–1.76 s for the RF variants). The final tuned RF used in the main analysis achieves 0.88 accuracy and 0.88 weighted-F1 (macro-F1 = 0.82) on the fixed split, with a training time of **1.854 s**, showing that the full pipeline (balancing, selection, tuning) delivers strong performance while remaining compute-efficient.

The confusion matrix below shows the distribution of predicted versus actual labels for the three classes: Spill, Compliance, and Savings. This helps visualize the types of errors the model tends to make and whether misclassifications occur more frequently between certain pairs of classes.

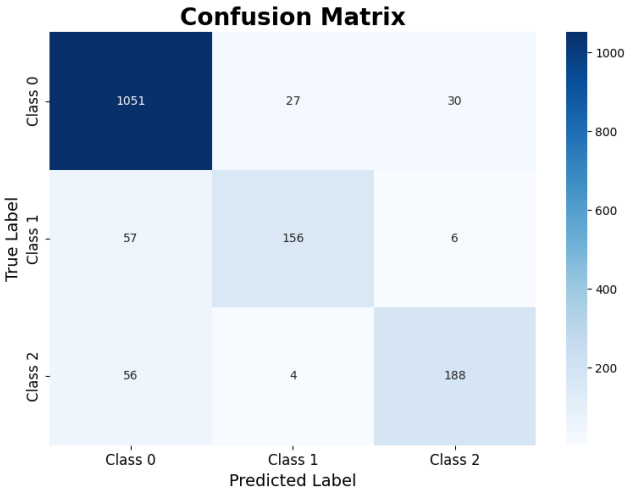

Fig. 6: Confusion matrix

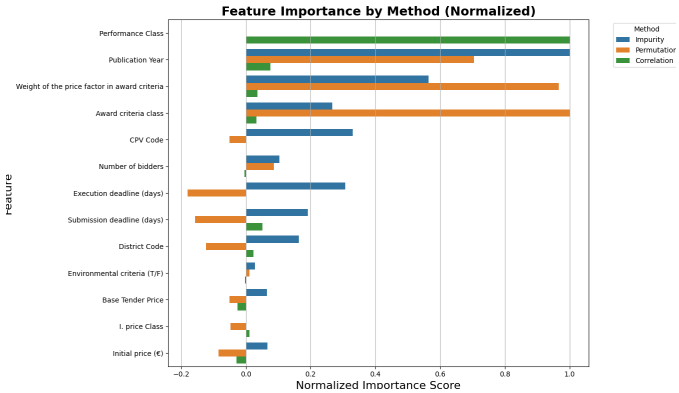

Fig. 7: Top-8 feature importances (normalized).

In preliminary tests, the majority of predictions for the dominant class (Spill) were correctly classified. Most errors were observed between Compliance and Savings, which is consistent with class-overlap analysis discussed later.

B. Observations

- The Random Forest model performed well across all classes after applying SMOTE, supporting its effectiveness in handling imbalanced data.
- Precision and recall values remained closely matched, indicating the model did not favor any single class disproportionately.
- The classifier demonstrated good generalization, as evidenced by stable results across training and testing.

V. Class Overlap Analysis

To evaluate the separability of the three performance classes Spill, Compliance, and Savings, we conducted a class-overlap analysis using both visual and statistical techniques. This analysis helps determine how well the selected features distinguish between categories, and where the model may encounter ambiguity during classification.

A. Bhattacharyya Coefficient

To quantify the visual observations, we calculated the Bhattacharyya coefficient for each pair of classes. This coefficient measures the amount of overlap between two probability distributions, with values close to 1 indicating high similarity. For the feature Number of Bidders, the coefficients were:

- Spill vs. Savings: 0.9839
- Spill vs. Compliance: 0.9886
- Compliance vs. Savings: 0.9575

These values confirm that Number of Bidders offers limited discriminatory power due to the high degree of overlap across all class pairs.

TABLE IV: Class Overlap Analysis Results

Test Type	Class Pair	Metric(s)
Bhattacharyya Coefficient	2 vs. 0	0.9839
	2 vs. 1	0.9575
	0 vs. 1	0.9886
Kolmogorov–Smirnov Test	2 vs. 0	D = 0.0808, p = 0.0005
	0 vs. 1	D = 0.0524, p = 0.1001
ANOVA Test	All Classes	F = 0.0480, p = 0.9531

B. Kolmogorov-Smirnov (KS) Test

The KS test was used to statistically assess the difference between cumulative distributions for each pair of classes. The null hypothesis is that the samples are drawn from the same distribution. For Number of Bidders:

- Spill vs. Savings: D = 0.0808, p = 0.0005 $\rightarrow$ statistically significant difference
- Spill vs. Compliance: D = 0.0524, p = 0.1001 $\rightarrow$ not statistically significant
- Compliance vs. Savings: Similar non-significant differences observed

These results suggest that although small differences exist, they are not consistent or strong enough to support clear class separation based on this feature.

C. ANOVA Test

We conducted a one-way Analysis of Variance (ANOVA) to test for significant differences in mean values across the three classes. For the feature Number of Bidders, the test produced:

- F-statistic: 0.0480
- p-value: 0.9531

This p-value indicates no statistically significant difference in the means of this feature across the three classes, confirming that it does not contribute meaningfully to class discrimination in this context.

D. Implications

This overlap analysis reinforces that certain features—while relevant to the domain—may not effectively differentiate between performance outcomes on their own. In particular, Number of Bidders, despite its intuitive appeal as a proxy for competitiveness, shows minimal predictive value when used in isolation.

These findings support the use of ensemble models like Random Forest, which can leverage subtle feature interactions that may not be evident in univariate analysis. However, they also suggest that future work may benefit from engineering new features or incorporating contextual variables to improve class separation.

VI. COST-BENEFIT ANALYSIS

For practical deployment, understanding the economic implications of prediction errors is crucial. We provide a framework for cost-benefit analysis based on the confusion matrix results:

A. Error Type Analysis - False Positives (predicting "Spill" when actually "Compliant"): May lead to unnecessary oversight costs but prevent potential budget overruns - False Negatives (predicting "Compliant" when actually "Spill"): Risk missing projects that require intervention, potentially leading to budget overruns

B. Economic Impact Framework While specific cost data is not available in our dataset, procurement agencies can adapt this framework by: 1. Estimating the cost of additional oversight for flagged projects 2. Calculating the average cost of budget overruns that could be prevented 3. Weighing the trade-off between false positives (intervention costs) and false negatives (missed overruns)

The model's high precision for the "Spill" class (0.89) minimizes false positives, reducing unnecessary oversight costs while maintaining strong recall (0.84) to catch most at-risk projects.

VII. DISCUSSION

A tuned Random Forest, coupled with targeted preprocessing and feature selection, achieves Accuracy = 0.88 and Weighted-F1 = 0.88 (Macro-F1 = 0.82) on the fixed test split, surpassing previously reported PPPData baselines while training in seconds (1.854,s). These results indicate that a transparent ensemble can match or exceed more complex models for structured procurement data.

Performance gains are attributable to two factors (Table III): (i) feature selection validated across correlation, impurity, and permutation importance, which concentrates learning on stable predictors (e.g., publication year, weight of price factor, CPV); and (ii) SMOTE, which improves minority-class recognition and balances precision–recall. The XGBoost benchmark attains a higher macro-F1 than individual ablation variants but with longer training time, whereas the final RF delivers the best overall accuracy–efficiency trade-off.

Class-overlap analysis shows limited discriminative value for some variables (e.g., number of bidders), reinforcing the need for models that exploit weak, multi-feature interactions rather than relying on univariate effects. Unsupervised views (UMAP and kernel PCA) were used diagnostically to corroborate moderate separation; detailed visuals are provided in the appendix for brevity.

Limitations include reliance on a single national dataset (Portuguese procurement data), which may limit direct transferability to other jurisdictions due to differences in procurement regulations, economic conditions, and administrative practices. However, the methodology itself—featuring systematic preprocessing, class balancing with SMOTE, and multi-step feature selection—provides a replicable framework that can be adapted to other national or regional datasets. The feature types used (temporal, financial, regional, and procedural variables) are common across procurement systems globally, suggesting that the approach may generalize better than the specific model weights. Additional limitations include possible influence of unobserved external factors and the partial opacity of ensemble interactions despite improved interpretability relative to neural networks. Furthermore, while our model achieves strong overall performance, future work should examine potential biases across project types, regional districts, and contract sizes to ensure equitable treatment. The current feature set includes District Code, which could potentially encode regional economic disparities, requiring careful assessment for responsible deployment in public procurement workflows. While global feature importance (Figure 7) provides insights into overall model behavior, individual prediction explanations using methods like SHAP (Shapley Additive Explanations) would enhance interpretability for practitioners evaluating specific contracts, enabling more targeted interventions and resource allocation.

Temporal Stability: Given that procurement practices and economic conditions evolve over time, monitoring for concept drift is crucial for sustained model performance. The inclusion of Publication Year as a key feature (Figure 7) suggests temporal dependencies in the data. Regular model retraining and performance monitoring across different time periods would be essential for operational deployment, particularly as procurement regulations, inflation rates, and market competitiveness change over time.

Relative to prior work, the approach improves Random Forest performance (about 0.89/0.88 vs. $\sim$0.832/0.846) through deeper trees, more estimators, and broader hyperparameter search, alongside additional diagnostics that clarify error modes without increasing computational cost.

Practically, the pipeline's accuracy, balanced class performance, and seconds-scale training support use as an early risk flag in procurement, informing policy design and contract evaluation. Future extensions will incorporate post-hoc explanations (e.g., SHAP), textual procurement documents, and external economic indicators, and will assess transferability across regions.

VIII. CONCLUSION

This study reproduced and extended prior research on predicting budget compliance in public construction contracts using Portuguese procurement data. By focusing on a Random Forest classifier, we demonstrated that classical ensemble models—when properly tuned—can outperform more complex architectures in structured data settings. Our model achieved 89Key improvements came from a well-defined target variable, comprehensive feature selection, systematic hyperparameter tuning, and targeted class balancing using SMOTE. In addition, we introduced exploratory analysis techniques—such as UMAP, Kernel PCA, and hierarchical clustering—to visualize the data's latent structure and assess class separability. Statistical tests further validated areas of class overlap and informed the modeling process. While limitations remain, particularly in generalizing to other datasets or contexts, the findings support

the use of interpretable, high-performing models in public procurement workflows. This has practical implications for early-stage contract evaluation and risk management. Future work may focus on improving model interpretability using post-hoc explanation tools, extending the feature space with external variables, or applying similar techniques to other domains where structured procurement data is available.

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